Project Information
Project Type: Team
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Research Title: Post-Disaster Hospital Simulation with Digital Twins and Adaptive Task Allocation

Problem Statement, Hypothesis, and Research Questions:

Problem Statement:

In emergency scenarios like post-earthquake hospital rescues, robotic agents must coordinate tasks such as triage, transport, and supply delivery in real time under uncertain, hazardous conditions. These scenarios demand not only rapid decision-making but also resilience to changing environments, incomplete information, and varying levels of agent reliability.

To address these challenges, digital twins—virtual replicas of physical agents and environments—can serve as a central coordination layer by enabling real-time monitoring, planning, and anomaly detection. However, effectively using digital twins for multi-agent task allocation requires optimizing across complex, dynamic constraints such as trust levels, blocked routes, and patient urgency.

This project investigates how quantum-enhanced graph optimization (via QUBO modeling) can improve digital twin-assisted coordination in a simulated rescue setting. The aim is to evaluate whether quantum-inspired methods can outperform classical task allocation strategies when trust, uncertainty, and environmental risk are encoded as graph attributes.

Hypotheses:

Trust-aware task assignment using quantum graph optimization will yield higher task completion rates (under time and safety constraints) than random or greedy allocation strategies in a simulated rescue scenario.

Digital twin models updated with real-time trust and environmental feedback will outperform static digital twin models in dynamic path planning tasks.

Research Questions:
RQ1: How can agent trust levels, task urgency, and environmental risks be modeled as a weighted graph suitable for QUBO formulation?

RQ2: Does using quantum-inspired optimization (e.g., Max-Cut or QAOA over a task-agent graph) improve task allocation efficiency in real-time rescue simulations?
RQ3: How does integrating a dynamic digital twin (updated with real-time data) affect the adaptability and performance of agent coordination in changing environments?

Literature Review

1. Safety Control of Service Robots with LLMs and Embodied Knowledge Graphs

- Authors: Yong Qi, Gabriel Kyebambo, Siyuan Xie, Wei Shen, Shenghui Wang, Bitao Xie, Bin He, Zhipeng Wang, Shuo Jiang
  Publication Year: 2024

Summary:
This paper proposes a new safety framework for service robots by combining Large Language Models (LLMs), Embodied Robotic Control Prompts (ERCPs), and Embodied Knowledge Graphs (EKGs). The goal is to improve how robots understand commands, break them down into safe task steps, and validate those steps before taking action. The ERCPs help LLMs generate clear, safety-focused task plans. The EKG acts like a dynamic fact-checker, making sure every step of the plan is safe and fits the current environment. The system was tested against LLM-only, KG-only, and LLM+KG systems. It achieved a 90% task success rate, fewer errors (3%), and fewer safety violations, showing stronger adaptability in changing environments. The study highlights the benefit of adding real-time validation and feedback loops to robot planning, but it notes challenges in scaling the knowledge graph and ensuring real-time performance in complex environments. This work informs how we might design Neo4j-based graphs that both represent knowledge and help robots act safely in dynamic rescue scenarios.

Relevance to Our Project:

This paper shows how a knowledge graph can do more than store data—it can actively validate plans for safety before they are executed. We can apply this idea by designing a Neo4j graph that checks task assignments and paths for hazards (e.g., blocked corridors) in our hospital rescue scenario, improving safety before sending tasks to the quantum optimizer.

Link: arXiv:2405.17846

2. UAV-UGV Cooperative Trajectory Optimization and Task Allocation for Medical Rescue Tasks in Post-Disaster Environments

Authors: Kaiyuan Chen, Wanpeng Zhao, Yongxi Liu, Yuanqing Xia, Wannian Liang, Shuo Wang
Publication Year: 2025

Summary:
This paper presents an integrated framework that coordinates unmanned aerial vehicles (UAVs) and unmanned ground vehicles (UGVs) for medical rescue operations in post-disaster environments. The authors address two main challenges: (1) how to assign tasks efficiently to a mixed fleet of UAVs and UGVs and (2) how to generate safe, efficient, and smooth trajectories through obstacle-filled environments.

Their solution combines an enhanced Genetic Algorithm (GA) for task allocation, Informed-RRT* for initial collision-free path planning, and Covariance Matrix Adaptation Evolution Strategy (CMA-ES) for optimizing task sequences and refining trajectories. This approach leverages the complementary strengths of UAVs (fast, agile) and UGVs (strong, high-capacity) to maximize mission efficiency.

The proposed method was tested through simulations in a realistic, earthquake-damaged urban scenario. Results showed that their system significantly outperformed simpler approaches. Specifically, it reduced mission completion time and total path length compared to:

- Random assignment (tasks given out randomly, no planning)
- K-Means clustering (tasks grouped by location without considering robot type or obstacles)
- Basic GA without advanced path optimization

Their method completed missions in as little as ~27 minutes, compared to ~100 minutes for random and K-Means methods, and showed strong scalability as the number of tasks and vehicles increased. The framework successfully generated collision-free, efficient routes and demonstrated that coordinated UAV-UGV planning can dramatically improve disaster response performance.

Relevance to Our Project:
We could use their random assignment results as a baseline for constructing our own tables of data, helping us compare more advanced trust-aware or quantum-enhanced task allocation strategies against a simple starting point.

Link: https://arxiv.org/pdf/2506.06136

3. Robotics Classification of Domain Knowledge Based on a Knowledge Graph for Home Service Robot Applications

Authors: Yiqun Wang, Rihui Yao, Keqing Zhao, Peiliang Wu, Wenbai Chen

Publication Year: 2024

Summary:

This study addresses the challenge of enabling home service robots to quickly and accurately recognize indoor environments and tools despite having limited labeled data. The authors propose a system combining a domain-specific knowledge graph with a lightweight deep learning model (MobileNetV3 enhanced by transfer learning). The knowledge graph captures relationships between rooms, tools, and their functions, supporting decision-making through reasoning at both attribute and instance levels.

The approach classifies indoor functional areas (e.g., kitchen, bedroom) using the Scene15 dataset and tools (e.g., ladle, hammer) using the UMD affordance dataset. Transfer learning is applied by pre-training on CIFAR-100 and fine-tuning for home contexts. Experimental results show that the system achieves high recognition accuracy (88% for scenes, nearly 100% for some tools) while maintaining fast inference speeds (63 ms/image). Compared to classical methods like ResNet or SVMs, the proposed method significantly improves both speed and accuracy, making it suitable for real-time home service robot applications.

Relevance to our project:

We similarly draw on this paper's approach of using a knowledge graph to connect rooms, objects, and their functions, helping our rescue robots reason about spaces (e.g., what tasks are likely in a damaged hospital wing).

Link: https://www.mdpi.com/2076-3417/14/24/11553

1. Modeling and Evaluating Trust Dynamics in Multi-Human Multi-Robot Task Allocation
   - Authors: Ike Obi, Ruiqi Wang, Wonse Jo, Byung-Cheol Min
   - Publication Year: 2025
   - Summary: This paper introduces the Expectation Confirmation Trust (ECT) model to analyze trust dynamics in multi-human, multi-robot (MH-MR) teams. The ECT model is grounded in the idea that trust plays a critical role in communications between MH-MR teams. These communications include things like effective coordination, adaptive decision making, and overall team performance in complex environments; the study demonstrates that incorporating trust models into task allocation algorithms improves task success rates and reduces completion times across various team configurations. The authors evaluated ECT against five other existing models and found that the ECT model is consistently better than the other models in performance. However, the study was limited to simulations and does not test with any real world deployment. Our work will extend on this study to test how well the ECT model does when tested with real deployment, not just simulations.
   - Link: arXiv:2409.16009arxiv.org+2arxiv.org+2arxiv.org+2

2. **[TIP: A Trust Inference and Propagation Model in Multi-Human Multi-Robot Teams]**

- Authors: Guo, Yang, & Shi
- Publication Year: 2023
- Summary: Summary: This paper introduces the Trust Inference and Propagation (TIP) model, which accounts for both direct and indirect human experiences when modeling trust in human-robot teams. Direct experience refers to firsthand interactions, while indirect experience involves observing or learning about others' interactions with robots. TIP defines trust as a weighted combination of these two types of experiences and uses a gradient-based optimization method to fit personalized trust curves for each human-robot pair. The model was validated through a 15-session user study involving 30 participants performing a drone-based search task. Results show that TIP significantly reduces trust prediction error (RMSE ≈ 0.06–0.08) compared to models relying only on direct experience (RMSE ≈ 0.08–0.11). The authors also provide a theoretical guarantee that the inferred trust values will converge over repeated interactions. Limitations include limited scalability (tested only on two humans and two drones) and reliance on a single modality (task outcomes only), excluding potentially informative data like latency or communication behaviors.
- Relevance to Our Project:
  - Digital-Twin Integration: TIP's trust update mechanism can be embedded into our digital twin knowledge graph to continuously track evolving trust in real-time.
  - Quantum-Enhanced Detection: TIP trust scores can feed into quantum-enhanced graph-based anomaly detection to flag potential manipulations (e.g., spoofed outcomes).
  - Adversarial Robustness: By simulating cyber-physical attacks in our digital twin system, we can observe TIP's behavior under adversarial conditions and adapt it to resist manipulation (e.g., through selective trust decay).

3. **[Attacking Digital Twins of Robotic Systems to Compromise Security and Safety]**

- Authors: Carr, Wang, Wang & Han
- Publication Year: 2022
- Summary: This paper investigates the security vulnerabilities of digital twin systems (DTS) in robotic platforms that rely on the Robot Operating System (ROS). The authors demonstrate how *Person-in-the-Middle* (PitM) attacks can compromise the

communication channel between a physical robot and its digital twin, leading to unsafe or malicious behaviors. Two case studies are presented: a TurtleBot 3 operating in Gazebo and a Universal Robot 10 in Unreal Engine. In both scenarios, attackers inject falsified velocity or joint-trajectory messages, effectively altering robot behavior without detection. The paper emphasizes that DTS architectures often inherit the security flaws of ROS 1, including the lack of built-in encryption and authentication. While the authors suggest mitigations such as topic encryption and migrating to ROS 2, they do not provide implementation or empirical evaluation of these defenses. Furthermore, the work does not incorporate anomaly detection or trust mechanisms to enable system-level resilience. Limitations include the exclusive focus on ROS 1 (excluding more secure middleware like ROS 2) and a lack of implemented or tested countermeasures.

- Relevance to Our Project:
    - We can embed TIP-style trust scores into the DTS so that sudden drops in trust (or unexpected message flows) automatically trigger alerts or fail-safe behaviors.
    - We can simulate these PitM scenarios in our digital-twin framework, measure resilience metrics under attack, and evaluate specific countermeasures rather than only proposing them.

4. Exploring the Synergy of Human-Robot Teaming, Digital Twins, and Machine Learning
    - Authors: Evan Langas, Muhammad Zafar, Filippo Sanfilippo
    - Publication Year: 2025
    - Summary: This paper discusses the integration of human-robot teaming with digital twin technology and machine learning. It highlights how these technologies can enhance perception, prediction, and decision-making in human-centric industrial systems. The methodology consists of a progressive framework: they start with exploring human-robot interaction (HRI), to human-robot collaboration (HRC), to physical HRC (pHRC), to true human-robot teaming (HRT). By integrating real time sensing, simulation, and intelligent decision-making, digital twins and machine learning can enhance robot perception, safety, and adaptability in dynamic industrial environments. They also discuss the ethical considerations of giving a robot autonomy, privacy, and how the designs have to be human-centric to make it more ethical. Some of the limitations include the fact that this paper is mostly conceptual; there is no real empirical data, no tests/experiments really conducted. Our work will extend on this by attempting to enhance digital twin technology using machine learning using some of the conceptual ideas they have came up with for enhancements.
    - Link: https://arxiv.org/abs/2410.18195
5. NeuronsGym: A Hybrid Framework and Benchmark for Robot Navigation With Sim2Real Policy Learning

- Authors: Haoran Li, Guangzheng Hu, Mingjun Ma, Yaran Chen, Dongbin Zhao
- Publication Year: 2024
- Summary: This paper presents NeuronsGym, a hybrid framework and benchmark for studying Sim2Real policy learning in robot navigation. It combines a Unity3D-based simulator with a real-world testbed using RoboMaster EP robots to enable efficient training and direct policy deployment. The simulator models robot dynamics and sensor behavior (LiDAR, odometer, camera) with realistic noise and anomalies to better match physical environments. A novel metric, Safety-Weighted Path Length (SFPL), is introduced to evaluate navigation safety by penalizing collisions, improving upon traditional metrics like SPL.NeuronsGym supports tunable motion and sensor parameters to study the Sim2Real gap under varying conditions. Experiments show that domain randomization methods like UDR and SimOpt improve real-world performance, especially under the PointGoal navigation task. LiDAR fidelity has a greater impact on navigation transfer than dynamics modeling, and increased robot speed worsens Sim2Real performance. The simulator is highly efficient, supporting fast, headless-mode training for large-scale experiments. While the framework emphasizes physical safety during navigation, it does not explore ethical issues like robot autonomy or privacy. A key limitation is the use of fixed environments. Our work will use this to help us know how to simulate an environment in Unity, and have the robot path around objects, and we can add onto this by adding dynamic pathing aspects using the quantum AI to allow for our robot to be able to move around dynamic objects (for more adaptability in real world applications).
- Link: https://ieeexplore.ieee.org/abstract/document/10750009

6. Learning Attribute Attention and Retrospect Location for Instance Object Navigation
   - Authors: Yanwei Zheng, Yaling Li, Changrui Li, Taiqi Zhang, Yifei Zou, Dongxiao Yu
   - Publication Year: 2025
   - Summary: This paper presents a novel cascade architecture to improve instance-level object navigation (ION), where agents must locate specific objects using fine-grained attributes like color, material, and reference cues. Current ION approaches often struggle with attribute confusion and weak memory of explored areas. To address these limitations, the authors introduce two main components: the Object-Attribute Attention Graph (OAAG) and the Objective Retrospect and Location Module (ORLM). OAAG enhances object discrimination through two sub-graphs: the Object-Aware Graph (OAG), which dynamically learns relationships among observed objects, and the Attribute-Attention Graph (AAG), which uses attention to focus on key distinguishing attributes. This helps the agent identify specific instances, even within the same category. ORLM improves memory and spatial reasoning through a

Back-tracker, which retains temporal and spatial object memory, and a Locator, which maps where targets were seen during exploration. Together, OAAG and ORLM provide stronger perception and memory capabilities.Integrated with an A3C reinforcement learning framework and tested in the AI2-THOR simulator, the model achieves state-of-the-art performance on Instance-Localization, Instance-Navigation, and Category-Localization tasks. The approach nearly doubles success rates over baseline methods. This will be useful to our project because we can use their improved ideas of ION to help us navigate dynamic simulated environments.

- Link: https://dl.acm.org/doi/10.1145/3706423

7. Personalized Instance-based Navigation Toward User-Specific Objects in Realistic Environments
   - Authors: Luca Barsellotti, Roberto Bigazzi, Marcella Cornia, Lorenzo Baraldi, Rita Cucchiara
   - Publication Year: 2024
   - Summary: This paper introduces Personalized Instance-based Navigation (PIN), a task where agents must locate a specific object instance (e.g., a child's favorite teddy bear) among similar items, without relying on contextual cues. Unlike traditional navigation tasks, PIN emphasizes fine-grained object recognition and personalized retrieval in complex, realistic environments.To support this, the authors present PInNED, a dataset built from Habitat-Matterport3D scenes augmented with photo-realistic 3D objects from Objaverse-XL. Each episode provides the agent with reference images on neutral backgrounds and textual descriptions, without environmental context. Objects are procedurally placed on furniture like beds and tables, and same-category distractors are included to increase difficulty. The dataset includes 338 unique instances across 18 categories, with over 865k training and 1.2k validation episodes. Target objects used in validation are unseen during training, requiring zero-shot generalization. PIN differs from existing tasks by focusing on movable, injected objects and emphasizing instance-level recognition rather than category-based search. Experiments show that modular navigation approaches outperform end-to-end models, which struggle with distinguishing similar objects and recovering from mistakes. Despite improvements, the task remains challenging, especially when distractors are present. This benchmark establishes a foundation for future work on personalized embodied navigation.This will help us with our project because we can use their ideas of how to identify specific objects, make it better (maybe through the use of upgraded knowledge graphs and quantum AI) and then apply it to our robot.
   - Link: https://arxiv.org/abs/2410.18195

Proposed Approach

High-Level Solution:

We are developing a trust-aware, quantum-optimized multi-agent coordination system for post-earthquake hospital rescue. The system integrates a dynamic knowledge graph (Neo4j-based) with quantum optimization and digital twin simulation to enable robots to adaptively allocate tasks and plan safe paths in real time.

Models & AI Methods

- Graph Neural Networks (GNNs) → (Planned) For learning dynamic trust and task relationships in the knowledge graph

- Quantum Optimization Algorithms

  - QUBO formulations → To encode task allocation & routing as binary optimization problems

  - QAOA (Quantum Approximate Optimization Algorithm) → For near-optimal task allocation and routing under trust and risk constraints

  - Quantum Annealing (D-Wave or simulators) → For solving QUBO models efficiently

- Autoencoders (Planned) → For anomaly detection in digital twin sensor data

Libraries / Tools

- Neo4j → Knowledge graph modeling of agents, tasks, environment, trust

- NetworkX → Synthetic graph generation for quantum optimization testing

- Qiskit / D-Wave Ocean SDK → Quantum optimization and simulation

- PyTorch → Planned for trust score learning and anomaly detection models

- Gymnasium → Reinforcement learning for agent behavior testing

Cybersecurity Context

- Digital Twin with Dynamic Trust Scores → Each robot's twin updates in real-time and monitors for anomalies (e.g., spoofed data, task failure)

- Threat Modeling → Simulated adversarial conditions (e.g., blocked hallways, sensor spoofing, communication jamming)

- Resilience Analysis → System performance under disruptions is measured (e.g., rerouting, reassigning tasks, trust recalibration)

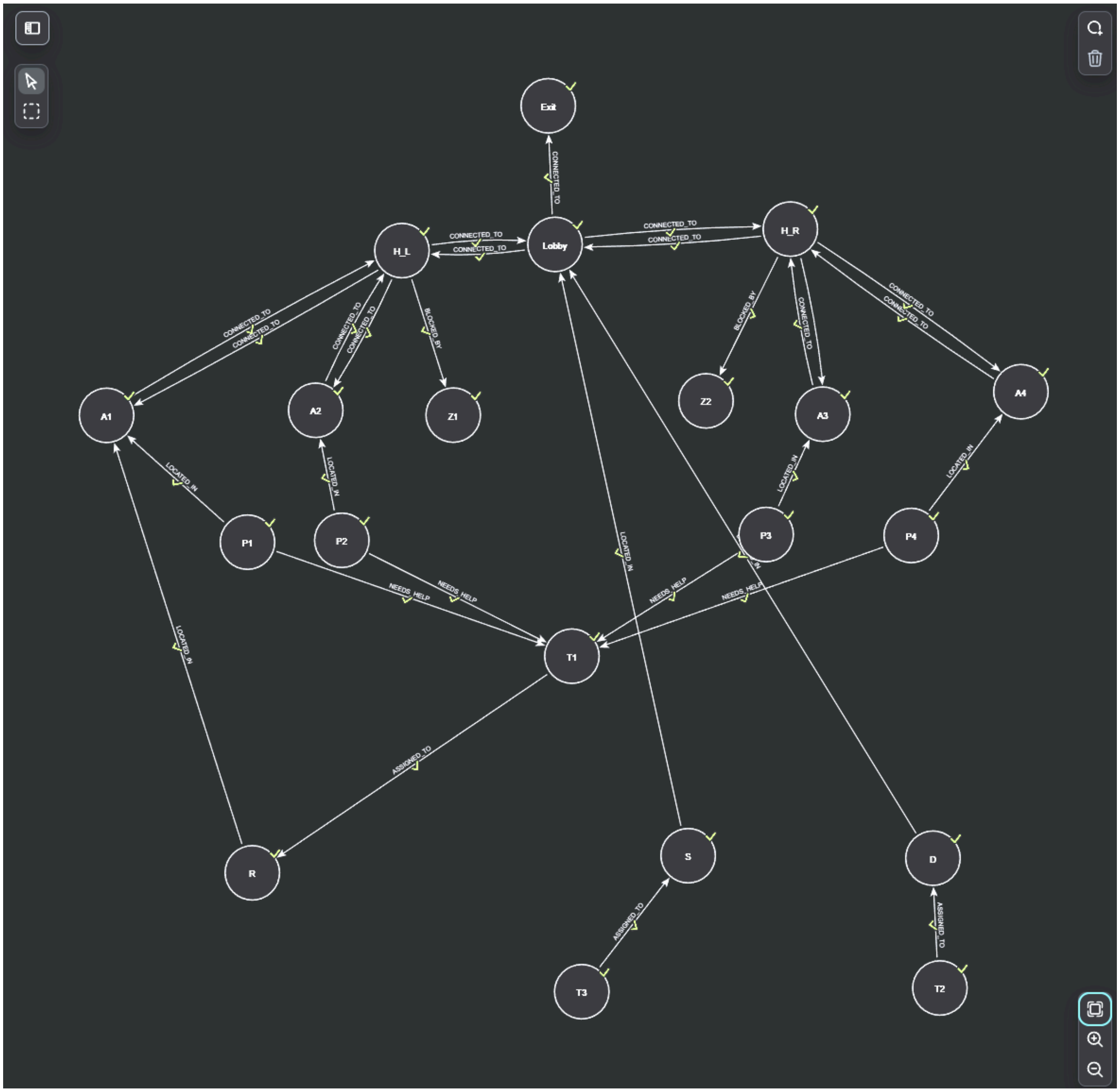
Preliminary Results

Graph Visualization Output

Using data from .csv files (agents.csv, patients.csv, tasks.csv, etc.), a Neo4j knowledge graph was created representing:

- Agents (robots, medics)
- Tasks (triage, supply delivery)
- Environmental entities (rooms, exits, hazards)
- Relationships (assigned tasks, blocked paths)

Graph, visualized with Neo4J:



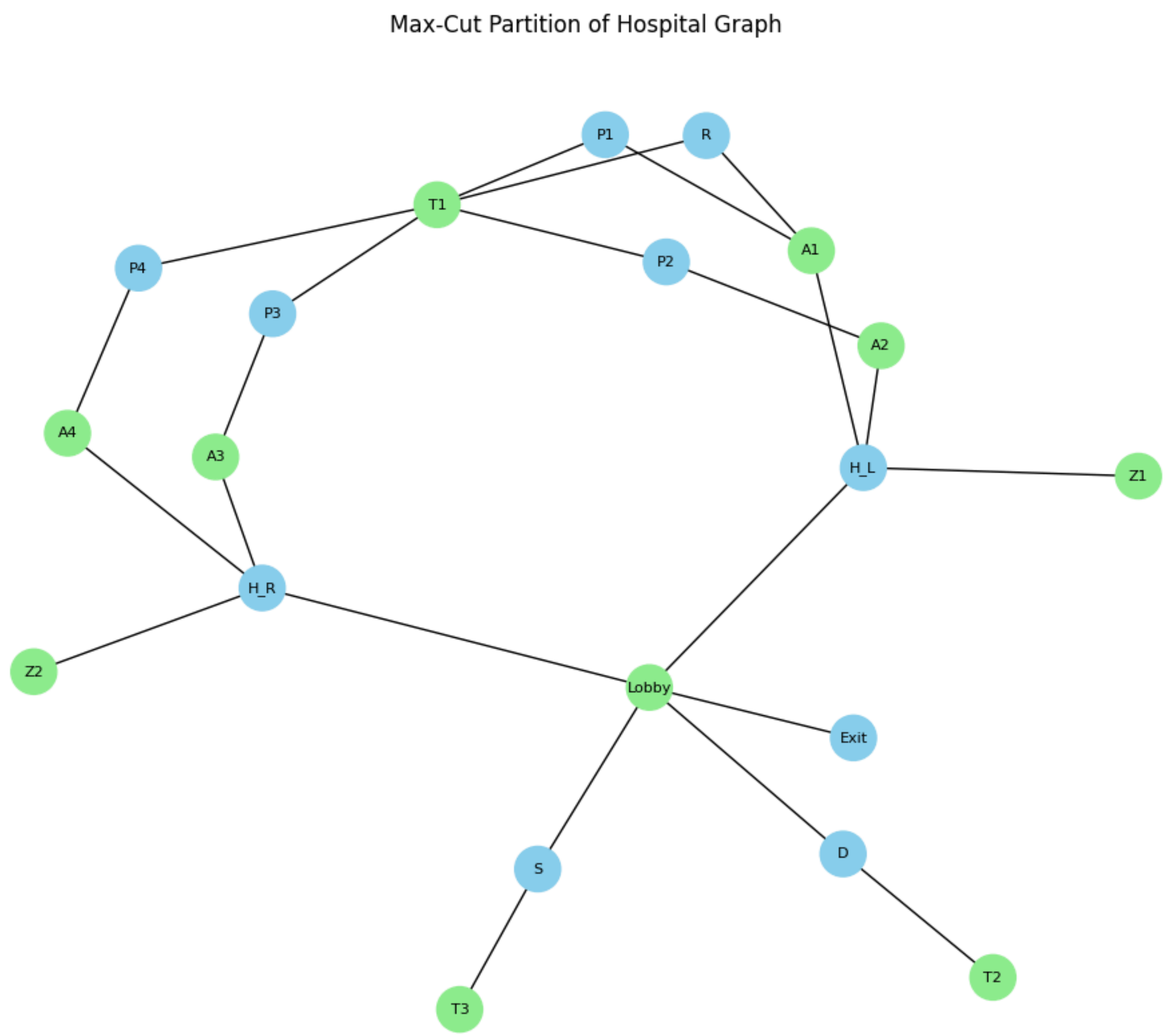
<u>Quantum Optimization using QUBO and the aforementioned graph:</u>

Initial efforts to simulate QUBO-based optimization using Qiskit failed due to dependency incompatibilities on Google Colab. As a workaround, the D-Wave Ocean SDK was used to implement a Max-Cut QUBO simulation on a small subset of the hospital graph.

Output:

Max-Cut Partition groups (two lists):

{'P2', 'H_R', 'R', 'Exit', 'H_L', 'S', 'D', 'P1', 'P4', 'P3'}



The partition is represented by the different color nodes (green is one group, blue is another group). This represents a logical partition of tasks or nodes that may indicate effective parallel assignment regions or routing clusters.

Challenges Encountered:

- Dependency conflicts: Qiskit optimization libraries (QAOA, QUBO) failed to import due to incompatible versions of qiskit.primitives.
- Simulation limits: D-Wave's local simulator (ExactSolver) handles only small-scale QUBO models.

High Level Proposed Research:

AI Models:

- Graph Neural Networks (GNNs) for modeling complex relationships between robots, environments, and tasks within dynamic knowledge graphs.

- Quantum Approximate Optimization Algorithm (QAOA) for real-time task allocation and trust-aware path planning.
- TIP and ECT trust updates embedded in a live knowledge graph (robots, tasks, sensors, humans).
- Autoencoders/anomaly detection models to identify compromised data flows or adversarial manipulation.

Tools and Libraries:

- Qiskit for QAOA and other quantum needs.
- PyTorch for developing and training classical machine learning models like GNNs or autoencoders.
- Hugging Face Transformers for integrating LLM -based reasoning into agent decisions or knowledge graph summarization.
- NetworkX/DGL for graph-based modeling of robots, environments, and trust relationships.
- Digital Twin Platforms (simulation, Unity3D specifically) to represent the physical-virtual interaction part of this project.

Cybersecurity Context:

Since we are focusing on implementing a digital twin for our project, these are the main cybersecurity concerns and focuses:

- Anomaly detection for identifying inconsistent or malicious behavior in sensor data or coordination plans.
- Trust modeling among multi-agent systems to simulate and adapt to potential deception or interference.
- Resilience analysis under adversarial scenarios (e.g., communication interference, spoofed commands) to test robustness of coordination protocols.

Experimental Design:

<u>Inputs</u>

- agents.csv, patients.csv, tasks.csv, environment_entities.csv, relationships.csv – structured hospital knowledge graph data
- Simulated disruptions (blocked routes, agent trust degradation, urgent patient tasks)
- Trust scores assigned to agents and updated dynamically
- What all csv files contain:

agents.csv

```
agentId,type,trust,trustTimestamp
R,robot,0.87,2025-06-20T16:00:00
S,staff,0.75,2025-06-20T16:01:00
D,drone,0.92,2025-06-20T16:02:00
```

environment_entities.csv

```
entityID,type
A1,room
A2,room
Lobby,room
A3,room
A4,room
H_L,blocked_hallway
H_R,blocked_hallway
Z1,hazardous_zone
Z2,hazardous_zone
Exit,safe_zone
```

patients.csv

```
patientID,priority
P1,high
P2,medium
P3,low
P4,high
```

tasks.csv

```
fromId,toId,type,assignedAt,eta,isBlocked,distance,requestedAt
P1,A1,LOCATED_IN,,,,,
P2,A2,LOCATED_IN,,,,,
P3,A3,LOCATED_IN,,,,,
P4,A4,LOCATED_IN,,,,,
R,A1,LOCATED_IN,,,,,
S,Lobby,LOCATED_IN,,,,,
D,Lobby,LOCATED_IN,,,,,
A1,H_L,CONNECTED_TO,,,false,7,
H_L,A1,CONNECTED_TO,,,false,7,
A2,H_L,CONNECTED_TO,,,false,5,
H_L,A2,CONNECTED_TO,,,false,5,
A3,H_R,CONNECTED_TO,,,false,5,
H_R,A3,CONNECTED_TO,,,false,5,
A4,H_R,CONNECTED_TO,,,false,7,
H_R,A4,CONNECTED_TO,,,false,7,
Lobby,H_L,CONNECTED_TO,,,true,5,
Lobby,H_R,CONNECTED_TO,,,true,5,
H_L,Lobby,CONNECTED_TO,,,true,5,
H_R,Lobby,CONNECTED_TO,,,true,5,
Lobby,Exit,CONNECTED_TO,,,false,10,
Z1,H_L,BLOCKED_BY,,,,,
Z2,H_R,BLOCKED_BY,,,,,
P1,T1,NEEDS_HELP,,,,,2025-06-20T15:58:00
P2,T1,NEEDS_HELP,,,,,2025-06-20T15:59:00
P3,T1,NEEDS_HELP,,,,,2025-06-20T16:02:00
P4,T1,NEEDS_HELP,,,,,2025-06-20T16:04:00
T1,R,ASSIGNED_TO,2025-06-20T16:05:00,5m,,,
T2,D,ASSIGNED_TO,2025-06-20T16:06:00,3m,,,
T3,S,ASSIGNED_TO,2025-06-20T16:07:00,7m,,,
```

tasks.csv

```
taskID,kind
T1,triage
T2,supply_delivery
T3,patient_extraction
```

Evaluation Metrics

Metric	Purpose
Task Success Rate	Measures percentage of tasks completed successfully
Latency	Time from task assignment to completion
Path Efficiency	Total distance/cost of agent routes compared to optimal
Trust Stability	Variance in agent trust scores over time

Baselines

Model Type	Description
Greedy Assignment	Agents take on the first available task
Random Assignment	Agents assigned tasks at random
Static Twin	Uses a non-updating digital twin model

Experimental Table

Experiment	Model	Dataset	Metric	Baseline	Expected Outcome
Exp1	Max-Cut QUBO	Rescue Graph	Task Success Rate	Greedy assignment	+10–15% success rate under trust constraints
Exp2	Trust Update Loop	Neo4j Twin + Relational Events	Trust Stability	Static Twin	Better stability and fewer conflicts
Exp3	Quantum Routing	Subgraph (agents + hazards)	Path Efficiency	BFS Routing	Shorter paths under dynamic hazards

Reproduce One or More SOTA Baselines:

Model Selection

- Name: Segment Anything Model (SAM), ViT-B variant
- Source: Barsellotti et al., "Personalized Instance-Based Navigation toward User-Specific Objects in Realistic Environments," NeurIPS 2024.
- Code repository: https://github.com/facebookresearch/segment-anything

Implementation Details

1. Environment:
   OS: macOS Sonoma
   Python: 3.9 inside a virtual environment (`.venv`)
   Hardware: CPU only (no GPU available)
2. Dependencies:

   pip install torch torchvision pillow

   pip install git+https://github.com/facebookresearch/segment-anything.git

Checkpoint Adaptation:

- The original ViT-H checkpoint (~2.6 GB) exceeds GitHub's 100 MB limit and crashes local downloads. I instead used the **ViT-B** checkpoint (~420 MB):

curl -L https://dl.fbaipublicfiles.com/segment_anything/sam_vit_b_01ec64.pth \ -o Week4-HandsOn/SegmentAnything/sam_vit_b.pth

- Removed the large file from Git history and updated `README.md` to fetch it at runtime.

  Demo Script (`demo.py`):

  - Loads `sam_vit_b.pth` via `sam_model_registry["vit_b"]`.
  - Reads a 320×240 JPEG (`test_image.jpg`) with PIL, converts to a NumPy array (`np.array(img)`) for compatibility with SAM's `.generate()` API.
  - Instantiates `SamAutomaticMaskGenerator` and prints the number of masks generated.

Initial Evaluation

- **Test image:** 320 × 240 sample.
- **Output:** 26 masks generated.
- **Runtime:** ~0.5 s per image on CPU (single-threaded).
- **Qualitative:** Masks outline major objects (desk, chair, computer) with moderate granularity.

Comparison to Original

	Original SAM ViT-H on GPU	Our SAM ViT-B on CPU
Model size	~2.6 GB	~420 MB
Speed	~0.05 s / image	~0.5 s / image
Mask count	Hundreds–thousands	26
Hardware	NVIDIA-class GPU	Intel-class CPU

- **Trade-off:** We accept fewer masks and slower speed to stay within resource and platform constraints.
- **Baseline value:** Provides a working segmentation component that can be integrated into our rescue-scenario pipeline; later replaced or augmented by a quantum-optimized, task-aware vision module.

Github of code:

https://github.com/josephkko/REU-2025-AIRobotics-Hands-On-Ko/tree/main/Week4-HandsOn/SegmentAnything